ANALYZE SALES OF VIDEO GAME

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Github- Shefali

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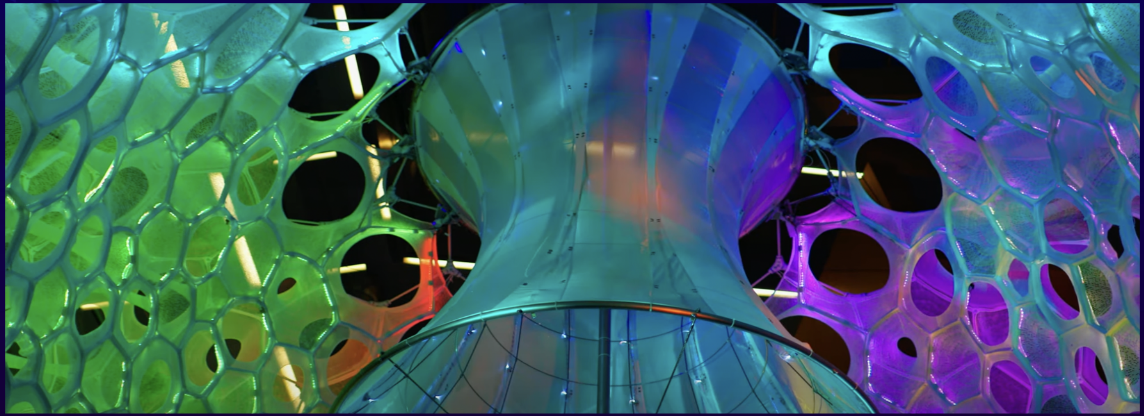
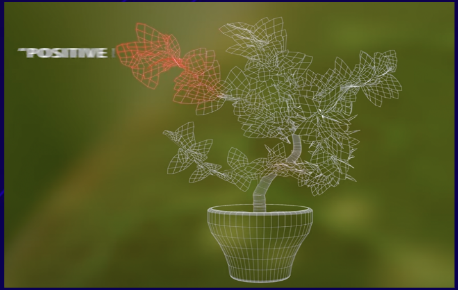
PERSONAL MOTIVATION

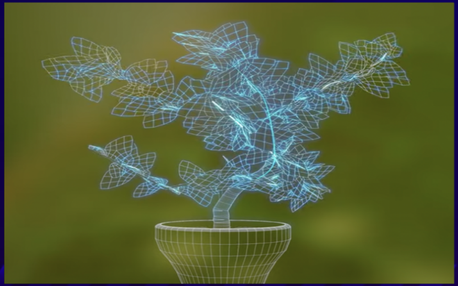
Power of Data Science – to train machines .
Inspired by Microsoft (ADA)
Plants responding to pollution by blinking LED lights.
Predict air quality using artificial intelligence.
Love for Games.



Positive response by plants



Negative response by plants



INTRODUCTION

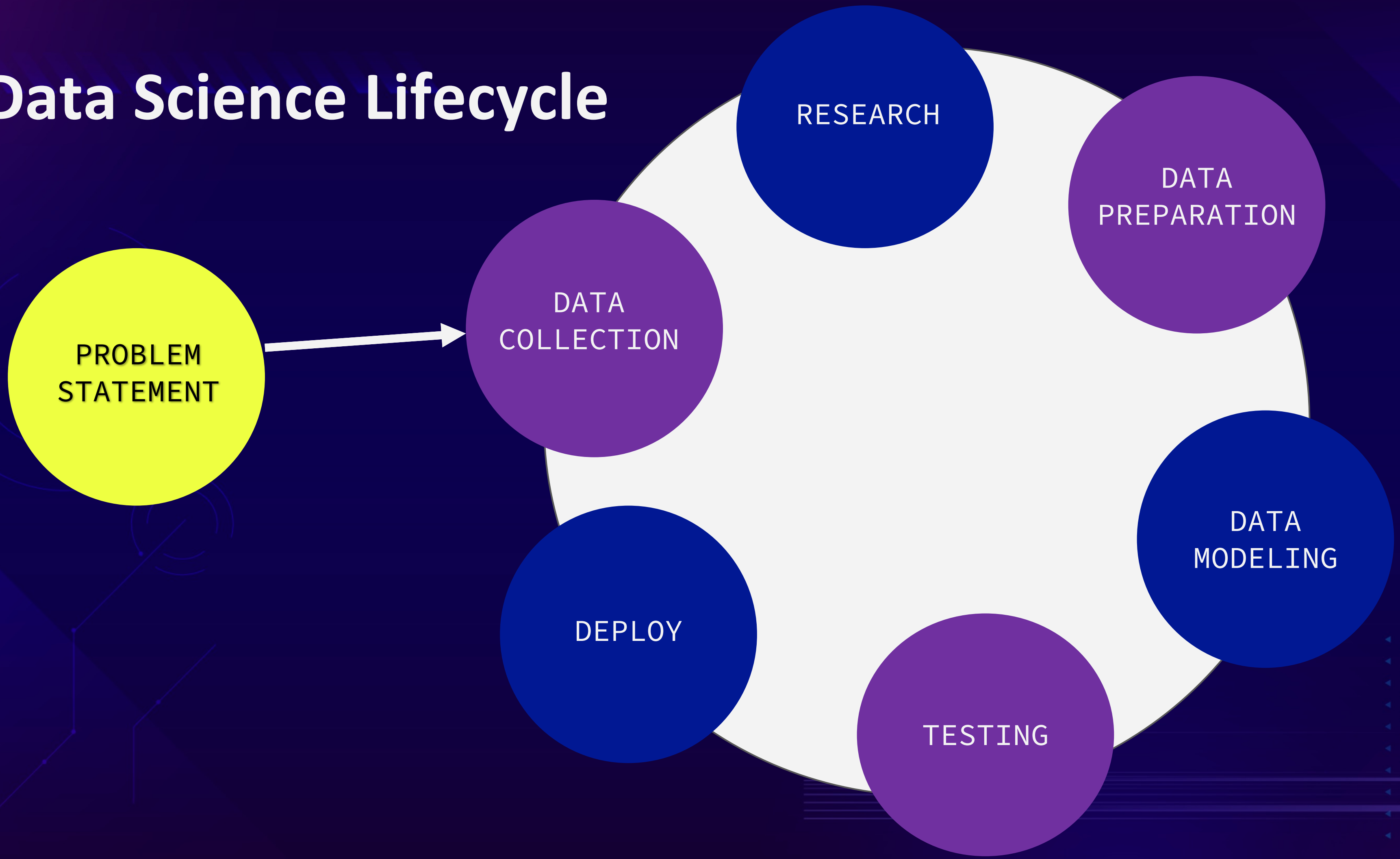
- The video game industry is the most trending industry globally which makes impact on growth of GDP.
- In the year 2020, USA generated $155 billion revenue by selling video games worldwide
- Recent studies shows that 1/3rd of global population widely play video games to reduce stress
- .

INTERESTING FACTS

- *Victor De Leon III* became youngest professional video gamer at the age of 11.
- *Carrie* made world-record by playing longest gaming session for 138 hours in California, USA.
- *“Gameboy”* is the first space game.
- Chun-li is the first female character in the video game.

- 10, 623 video games released during the pandemic year 2020.
- Approx. 1.1 Million video games delivered worldwide till date.
- *“Minecraft”* is the most played game which sold 33 billion units till date.
- ***“Tennis for Two”*** is the first video game developed by William Higinbotham.
- ***“Super Color Runner”*** high-priced video game for $200.

Data Science Lifecycle



PHASE 1: BUSINESS PROBLEM

- Video game companies need to identify which video game genre(category) is trending globally.
- Thus, finding best algorithm for prediction becomes crucial task for Data Scientist.

?

RESEARCH QUESTION

Which video game genre(category) is trending the most throughout the year?

PHASE 2: DATA COLLECTION

Dataset link

https://www.kaggle.com/gregorut/videogamesales

Rows and columns

16600 rows and 11 columns.

DESCRIPTION

- Consist of video games list based on individual ranking.
- From year 1980-2020.
- On which platform it was released.
- Sales of different regions.

PARAMETER

- Platform(Wii, Xbox 360, etc.), Year (1980-2020), Region (NA, EU, JP, Global), Publisher(Nintendo, etc.), Name (Super Mario, Tetris, etc.), Genre(Racing, Puzzle, etc.)

DATASET SAMPLE

vgsales

Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
1	Wii Sports	Wii	2006	Sports	Nintendo	41.49	29.02	3.77	8.46	82.74
2	Super Mario Bros.	NES	1985	Platform	Nintendo	29.08	3.58	6.81	0.77	40.24
3	Mario Kart Wii	Wii	2008	Racing	Nintendo	15.85	12.88	3.79	3.31	35.82
4	Wii Sports Resort	Wii	2009	Sports	Nintendo	15.75	11.01	3.28	2.96	33
5	Pokemon Red/Pokemon Blue	GB	1996	Role-Playing	Nintendo	11.27	8.89	10.22	1	31.37
6	Tetris	GB	1989	Puzzle	Nintendo	23.2	2.26	4.22	0.58	30.26
7	New Super Mario Bros.	DS	2006	Platform	Nintendo	11.38	9.23	6.5	2.9	30.01
8	Wii Play	Wii	2006	Misc	Nintendo	14.03	9.2	2.93	2.85	29.02
9	New Super Mario Bros. Wii	Wii	2009	Platform	Nintendo	14.59	7.06	4.7	2.26	28.62
10	Duck Hunt	NES	1984	Shooter	Nintendo	26.93	0.63	0.28	0.47	28.31

PHASE 3: RESEARCH (LITERATURE REVIEW)

A. Aziz, S. Ismail, M.F. Othman and A. Mustapha

The first research used random forest to study about the factors which are increasing the sales of video games.

S. Göring , R. Steger, R. Rao Ramachandra Rao and A. Raake

The second research chosen KNN to evaluate the dataset of video game genre graphical properties.

E.N Bailey and K. Miyata

The third research after suggested KNN to evaluate the multiplayer games dataset.

Phase 4: Data Preparation

FIRST 5 ROWS

```
[176] from google.colab import drive
      drive.mount("/content/drive")
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
[177] df = pd.read_csv('/content/drive/My Drive/CS668/vgsales.csv')
```

```
df.head()
```

	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
0	1.0	Wii Sports	Wii	2006.0	Sports	Nintendo	41.49	29.02	3.77	8.46	82.74
1	2.0	Super Mario Bros.	NES	1985.0	Platform	Nintendo	29.08	3.58	6.81	0.77	40.24
2	3.0	Mario Kart Wii	Wii	2008.0	Racing	Nintendo	15.85	12.88	3.79	3.31	35.82
3	4.0	Wii Sports Resort	Wii	2009.0	Sports	Nintendo	15.75	11.01	3.28	2.96	33
4	5.0	Pokemon Red/Pokemon Blue	GB	1996.0	Role-Playing	Nintendo	11.27	8.89	10.22	1.00	31.37

```
[179] df.tail()
```

	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
16594	16597.0	Men in Black II: Alien Escape	GC	2003.0	Shooter	Infogrames	0.01	0.00	0.0	0.0	0.01
16595	16598.0	SCORE International Baja 1000: The Official Game	PS2	2008.0	Racing	Activision	0.00	0.00	0.0	0.0	0.01
16596	16599.0	Know How 2	DS	2010.0	Puzzle	7G//AMES	0.00	0.01	0.0	0.0	0.01
16597	16600.0	Spirits & Spells	GBA	2003.0	Platform	Wanadoo	0.01	0.00	0.0	0.0	0.01
16598	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	

LAST 5 ROWS

DATA CLEANING

CHANGING DATA TYPE

```
df.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 16291 entries, 0 to 16597
Data columns (total 11 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Rank          16291 non-null  float64
 1   Name          16291 non-null  object
 2   Platform      16291 non-null  object
 3   Year          16291 non-null  float64
 4   Genre         16291 non-null  object
 5   Publisher     16291 non-null  object
 6   NA_Sales      16291 non-null  float64
 7   EU_Sales      16291 non-null  float64
 8   JP_Sales      16291 non-null  float64
 9   Other_Sales   16291 non-null  float64
 10  Global_Sales  16291 non-null  object
dtypes: float64(6), object(5)
memory usage: 1.5+ MB
```

```
[76] df["Global_Sales"] = df['Global_Sales'].astype(float)

df.info() #convert object dtype to float

<class 'pandas.core.frame.DataFrame'>
Int64Index: 16291 entries, 0 to 16597
Data columns (total 11 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Rank          16291 non-null  float64
 1   Name          16291 non-null  object
 2   Platform      16291 non-null  object
 3   Year          16291 non-null  float64
 4   Genre         16291 non-null  object
 5   Publisher     16291 non-null  object
 6   NA_Sales      16291 non-null  float64
 7   EU_Sales      16291 non-null  float64
 8   JP_Sales      16291 non-null  float64
 9   Other_Sales   16291 non-null  float64
 10  Global_Sales  16291 non-null  float64
dtypes: float64(7), object(4)
memory usage: 1.5+ MB
```

```
df.isnull().sum().to_frame()
```

	0
Rank	1
Name	1
Platform	1
Year	272
Genre	1
Publisher	59
NA_Sales	1
EU_Sales	1
JP_Sales	1
Other_Sales	1
Global_Sales	0

```
[181] print('The shape of the dataset:', df.shape)

The shape of the dataset: (16599, 11)
```

```
#drop all rows that have any Null values
df.dropna(inplace=True)

df.isnull().sum().to_frame()
```

	0
Rank	0
Name	0
Platform	0
Year	0
Genre	0
Publisher	0
NA_Sales	0
EU_Sales	0
JP_Sales	0
Other_Sales	0
Global_Sales	0

```
[61] print('The shape of the dataset:', df.shape)

The shape of the dataset: (16291, 11)
```

CHECK IF THERE IS ANY NULL VALUE

DROP NULL VALUES

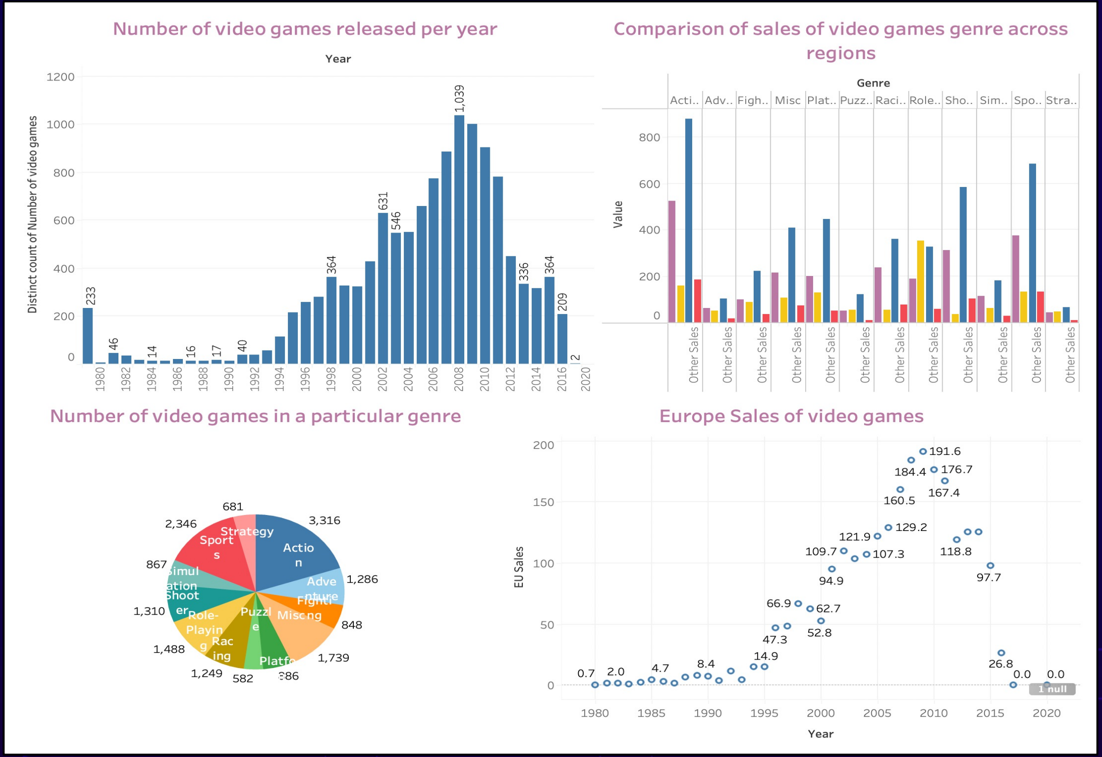
DISTRIBUTE THE DATASETS IN NUMERICAL FEATURES, CATEGORICAL FEATURES

```python
import numpy as np
# use this for datasets with more columns, to print all columns
np.set_printoptions(threshold=np.inf)

# This prints the column labels of the dataframe
print('All dataset columns:')
print(df.columns.values)

# This prints the column labels of the features identified as numerical
print('Numerical columns:')
print(df.select_dtypes(include=np.number).columns.values)

# This prints the column labels of the features identified as numerical
print('Categorical columns:')
print(df.select_dtypes(include='object').columns.values)
```

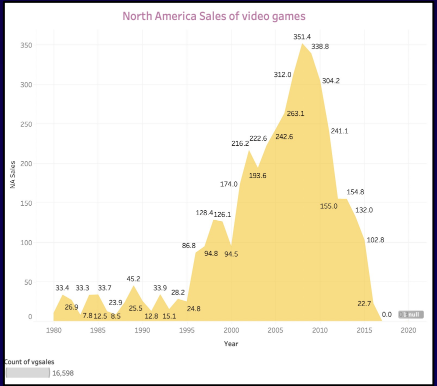
```
All dataset columns:
['Rank' 'Name' 'Platform' 'Year' 'Genre' 'Publisher' 'NA_Sales' 'EU_Sales'
 'JP_Sales' 'Other_Sales' 'Global_Sales']
Numerical columns:
['Rank' 'Year' 'NA_Sales' 'EU_Sales' 'JP_Sales' 'Other_Sales']
Categorical columns:
['Name' 'Platform' 'Genre' 'Publisher' 'Global_Sales']
```

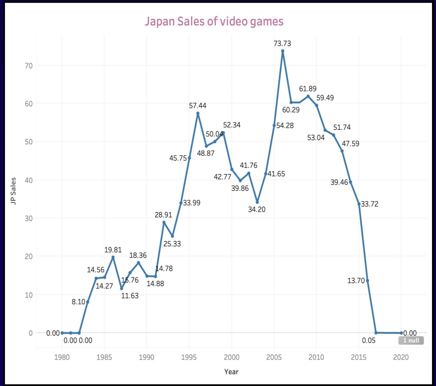
DATA VISAULIZATION using Tableau

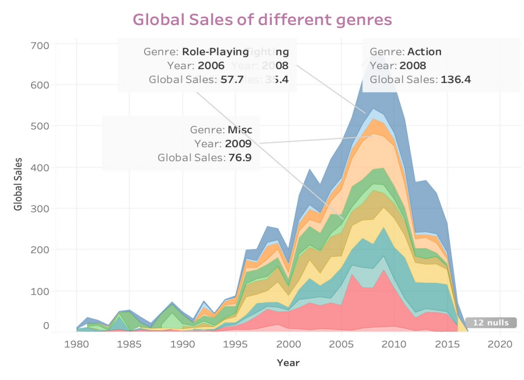
Link to view tableau dashboard

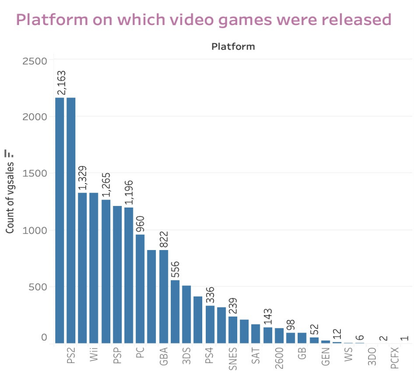


Cont.

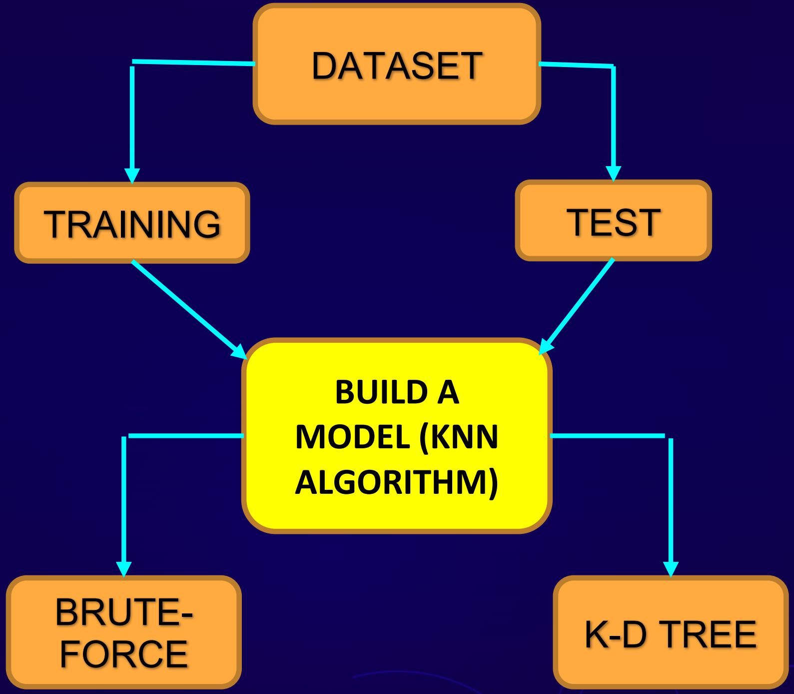

North America Sales of video games
NA Sales
Year
351.4
338.8
312.0
304.2
263.1
242.6
241.1
222.6
216.2
193.6
174.0
155.0
154.8
132.0
128.4
126.1
102.8
94.8
94.5
86.8
45.2
33.4
33.3
33.7
33.9
28.2
26.9
25.5
24.8
23.9
22.7
15.1
12.8
12.5
8.5
7.8
0.0
1 null
Count of vgsales
16,598


Japan Sales of video games
JP Sales
Year
73.73
61.89
60.29
59.49
57.44
54.28
53.04
52.34
51.74
50.04
48.87
47.59
45.75
42.77
41.76
41.65
39.86
39.46
34.20
33.99
33.72
28.91
25.33
19.81
18.36
15.76
14.88
14.78
14.56
14.27
13.70
11.63
8.10
0.05
0.00
1 null


Global Sales of different genres
Global Sales
Year
Genre: Role-Playing
Year: 2006
Global Sales: 57.7
Genre: Action
Year: 2008
Global Sales: 136.4
Genre: Misc
Year: 2009
Global Sales: 76.9
12 nulls

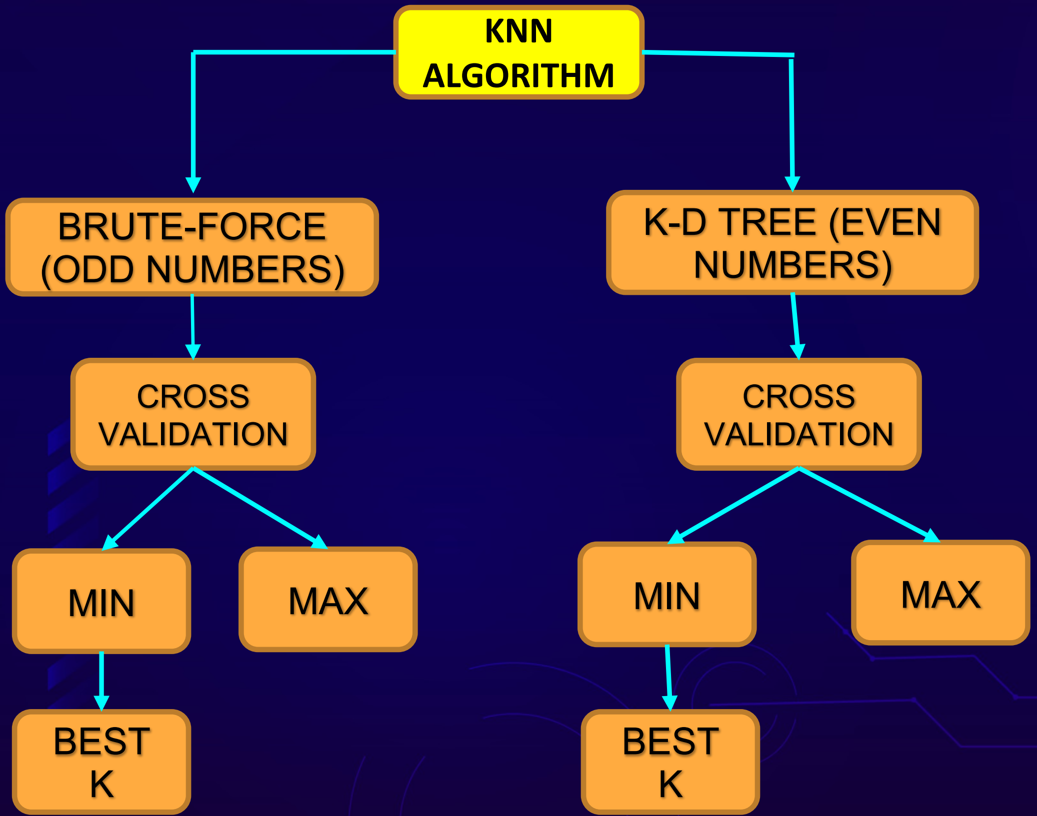

Platform on which video games were released
Platform
Count of vgsales
2,163
1,329
1,265
1,196
960
822
556
336
239
143
98
52
12
6
2
1
PS2
Wii
PSP
PC
GBA
3DS
PS4
SNES
SAT
2600
GB
GEN
WS
3DO
PCFX

PHASE 5: DATA MODELING



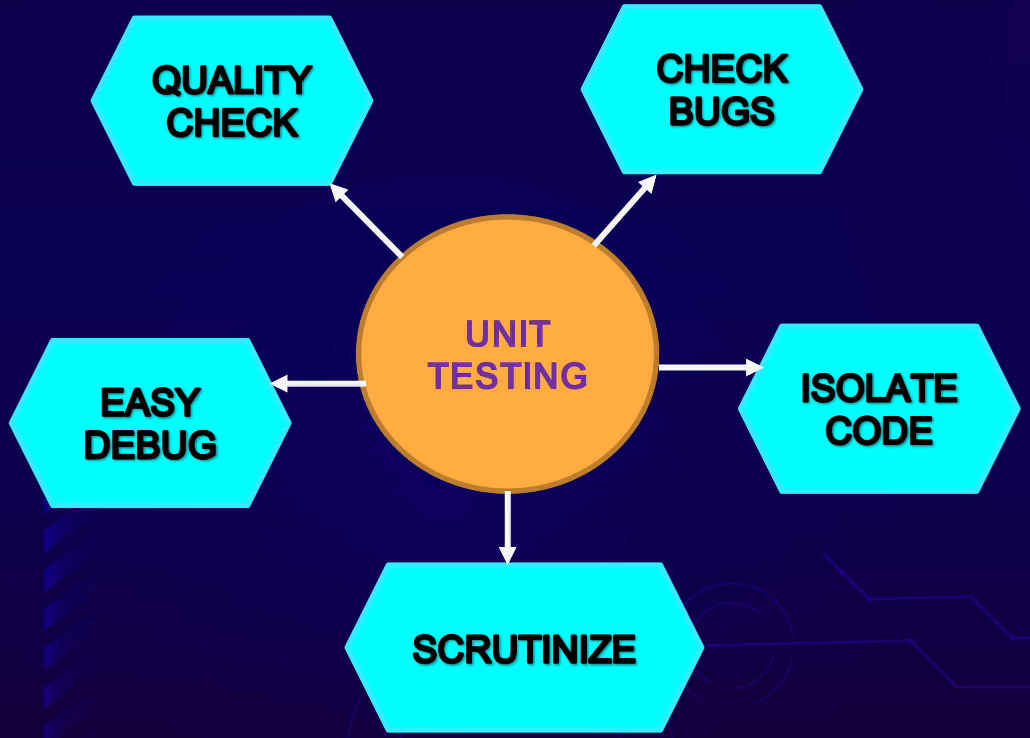
METHODOLOGY

Cont.



KNN ALGORITHM
BRUTE-FORCE (ODD NUMBERS)
K-D TREE (EVEN NUMBERS)
CROSS VALIDATION
CROSS VALIDATION
MIN
MAX
MIN
MAX
BEST K
BEST K

PHASE 6: TESTING(UNIT TEST)



DEMO

CS668 - Google Drive | vgunittest_success_HTMLrepo

colab.research.google.com/drive/1p-gxKdVRDbmZSdwtCyQUJGtu4pPElkr9#scrollTo=jPZmNfa1UNrZ

vgunittest_success_HTMLreport.ipynb

File Edit View Insert Runtime Tools Help

Comment Share

+ Code + Text

RAM Disk Editing

```
pip install html-testRunner
```

```
import numpy as np
import pandas as pd
import unittest
import matplotlib
import matplotlib.pyplot as plt
import HtmlTestRunner

import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import mean_squared_error
from sklearn.metrics import mean_absolute_error, accuracy_score
from sklearn.model_selection import cross_val_score
from sklearn.metrics import r2_score
import warnings
warnings.filterwarnings("ignore")
```

[24]
```
from google.colab import drive
drive.mount("/content/drive")
```

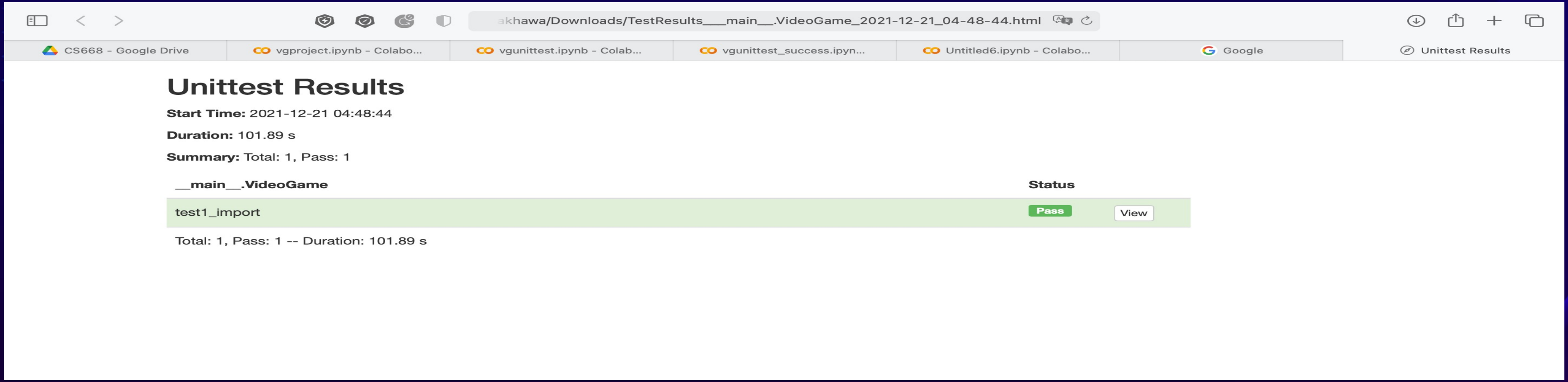
```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

[25]
```
class VideoGame(unittest.TestCase):

  def setUp(self):
    self.df = pd.read_csv('/content/drive/My Drive/CS668/vgsales.csv')


  def test1_import(self):
```

13s completed at 2:09 AM

UNITTEST REULT

```
OK (101.888098)s

----------------------------------------------------------------------
Ran 1 test in 0:01:41

OK



Generating HTML reports...
drive/My Drive/CS668/TestResults___main__.VideoGame_2021-12-21_04-48-44.html
<unittest.main.TestProgram at 0x7f4d2e0d6550>
```



RESULTS

ERROR VALUE

Actual Error Value		
ODD	3	0.01
	49	0.03
EVEN	2	0.01
	48	0.03

ERROR

Error in range (0,50)		
	Min	Max
ODD	3	49
EVEN	2	48

ACCURACY RATE

Algorithm	Value of K	Accuracy Rate Predicted
Brute -force	ODD K= 3	98.64%
K-d tree	EVEN K = 2	98.89%

CLASSIFICATION REPORT

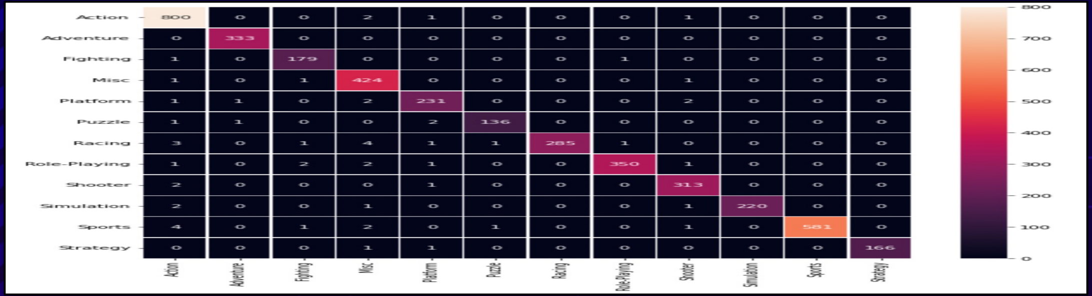
```
classification_report using kd-tree
              precision    recall  f1-score   support

      Action       0.99      1.00      0.99       804
   Adventure       1.00      1.00      1.00       333
    Fighting       0.98      0.99      0.99       181
        Misc       0.98      1.00      0.99       427
    Platform       0.96      0.98      0.97       237
      Puzzle       0.99      0.99      0.99       140
      Racing       1.00      0.97      0.98       296
Role-Playing       0.99      0.99      0.99       357
     Shooter       0.99      0.99      0.99       316
  Simulation       1.00      0.99      0.99       224
      Sports       1.00      0.98      0.99       590
    Strategy       1.00      0.99      0.99       168

    accuracy                           0.99      4073
   macro avg       0.99      0.99      0.99      4073
weighted avg       0.99      0.99      0.99      4073
```

CONFUSION MATRIX

	Action	Adventure	Fighting	Misc	Platform	Puzzle	Racing	Role-Playing	Shooter	Simulation	Sports	Strategy
Action	800	0	0	1	1	0	0	2	0	0	0	0
Adventure	0	333	0	0	0	0	0	0	0	0	0	0
Fighting	1	0	179	0	0	0	0	1	0	0	0	0
Misc	0	0	0	426	0	0	0	0	1	0	0	0
Platform	0	1	1	2	233	0	0	0	0	0	0	0
Puzzle	0	0	0	1	1	138	0	0	0	0	0	0
Racing	4	0	1	2	1	1	287	0	0	0	0	0
Role-Playing	0	0	1	2	0	0	1	352	1	0	0	0
Shooter	2	0	0	0	1	0	0	0	313	0	0	0
Simulation	2	0	0	0	0	0	0	1	0	221	0	0
Sports	2	0	0	2	4	0	0	0	1	1	580	0
Strategy	0	0	0	0	2	0	0	0	0	0	0	166

CONT.

HEATMAP



CONCLUSION

This project compares the performance of different models on the video game dataset. In this we have compared within the algorithm to predict which is the better

BEST K FOR ODD & EVEN NO.

K = 3 , K =2 as it give min value error

BEST MODEL

K-D TREE as highest accuracy rate of 98.89%

References

- A. Aziz, S. Ismail, M.F. Othman and A. Mustapha, “Empirical Analysis on Sales of Video Games: A Data Mining Approach”, (2018), Journal of Physics Conference Series 1049(1):012086. www.researchgate.net/Empirical_Analysis_on_Sales_of_Video_Games_A_Data_Mining_Approac
- S. Göring , R. Steger, R. Rao Ramachandra Rao and A. Raake, "Automated Genre Classification for Gaming Videos," (2020) IEEE 22nd International Workshop on Multimedia Signal Processing (MMSP), 2020, pp. 1-6, doi: 10.1109/MMSP48831.2020.9287122. https://ieeexplore.ieee.org/document/9287122
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- R. Crandall and G. Sidak, “Video Games: Serious Business for America's Economy”, (2007), https://www.researchgate.net/publication/228230702_Video_Games_Serious_Business_for_America’s
- X. Li, "Towards Factor-oriented Understanding of Video Game Genres using Exploratory Factor Analysis on Steam Game Tags," (2020), IEEE International Conference on Progress in Informatics and Computing (PIC), 2020, pp. 207-213, doi: 10.1109/PIC50277.2020.9350753. https://ieeexplore.ieee.org/document/9350753.

THANKS

Does anyone have any questions?

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